

OS JACKFRUIT

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1.Screenshot 1 — Multi-container supervision

Description:

Two containers (alpha, beta) running simultaneously under one supervisor process

TERMINAL-1

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs-base  
[sudo] password for heemadhawala:  
open /dev/container_monitor: No such file or directory  
Supervisor ready on /tmp/mini_runtime.sock  
█
```

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs-base  
[sudo] password for heemadhawala:  
open /dev/container_monitor: No such file or directory  
Supervisor ready on /tmp/mini_runtime.sock  
Received start for alpha, spawned pid 4384  
Received start for beta, spawned pid 4411  
█
```

TERMINAL -2

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha rootfs-base /bin/sh  
[sudo] password for heemadhawala:  
started pid 4384  
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta rootfs-base /bin/sh  
started pid 4411  
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ █
```

COMMANDS sudo ./engine supervisor rootfs-base
(TERMINAL -1) sudo ./engine start alpha rootfs-base /bin/sh
(TERMINAL-2) sudo ./engine start beta rootfs-base
/bin/sh(TERMINAL-2)

2.Screenshot 2 — Metadata tracking

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
ID          PID      STATE    SOFT_MIB HARD_MIB
beta       4411    running     40       64
alpha      4384    running     40       64
```

Description:

Output of **engine ps** showing container ID, PID, state, and memory limits.

COMMANDS

sudo ./engine ps

3.Screenshot 3 — Bounded-buffer logging

```
heemadhawala@debian: ~/OS-Jackfruit/boilerplate$ export PS1="jyothika@os-jackfruit: (~)$"
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs-base
[sudo] password for heemadhawala:
open /dev/container_monitor: No such file or directory
Supervisor ready on /tmp/mini_runtime.sock
Received start for alpha, spawned pid 4384
Received start for beta, spawned pid 4411
Received start for log1, spawned pid 4431
```

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start log1 rootfs-base /bin/date
sudo ./engine logs log1
started pid 4431
Tue Apr 14 11:13:09 UTC 2026
Wed Apr 15 10:13:45 UTC 2026
```

Description:

Log output captured through the producer–consumer logging pipeline.

COMMANDS

sudo ./engine start log1 rootfs-base /bin/date sudo

./engine logs log1

4.Screenshot 4 — CLI and IPC

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs-base
[sudo] password for heemadhawala:
open /dev/container_monitor: No such file or directory
Supervisor ready on /tmp/mini_runtime.sock
Received start for alpha, spawned pid 4384
Received start for beta, spawned pid 4411
Received start for log1, spawned pid 4431
Received start for t2, spawned pid 4452
█

jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start t2 rootfs-base /bin/sh
sudo ./engine stop t2
started pid 4452
sent SIGTERM to 4452
```

Description:

A CLI **stop** command sent to the supervisor via UNIX socket and handled correctly.

COMMANDS

```
sudo ./engine start t2 rootfs-base /bin/sh sudo
./engine stop t2
```

5.Screenshot 5 — Soft-limit warning

Description:

Kernel logs showing warning when container exceeds soft memory limit.

```
[ 2292.702701] [container_monitor] Registering container soft pid=6590 soft=11
[ 2296.627629] [container_monitor] SOFT LIMIT container=soft pid=6590 rss=4243
```

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start soft rootfs-base ./memory_hog
started pid 5735
```

COMMANDS

```
sudo ./engine start soft rootfs-base ./memory_hog sudo
dmesg | grep container_monitor
```

6.Screenshot 6 — Hard-limit enforcement

Description:

Kernel logs showing process termination when hard memory limit is exceeded.

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor
```

```
[ 2299.702452] [container_monitor] HARD LIMIT container=soft pid=6590 rss=67624960 limit=67108864
```

7.Screenshot 7 — Scheduling experiment

Description:

Two containers running CPU workloads with different priorities showing scheduling behavior.

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine start high rootfs-base ./cpu_hog --nice -5
sudo ./engine start low rootfs-base ./cpu_hog --nice 10
started pid 6509
```

```
bash: hog: command not found
started pid 6517
bash: hog: command not found
```

```
jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo ./engine logs high
sudo ./engine logs low
```

```
cpu_hog alive elapsed=1 accumulator=11743365965695951113
cpu_hog alive elapsed=2 accumulator=12705706947733282540
cpu_hog alive elapsed=3 accumulator=6536592348963765490
cpu_hog alive elapsed=4 accumulator=2748803756655716212
cpu_hog alive elapsed=5 accumulator=5471689385478181812
cpu_hog alive elapsed=6 accumulator=1879574134951606724
cpu_hog alive elapsed=7 accumulator=16439616552241313599
cpu_hog alive elapsed=8 accumulator=14985577974824870241
cpu_hog alive elapsed=9 accumulator=11701429646925013122
cpu_hog done duration=10 accumulator=12397476453989931003
```

```
cpu_hog alive elapsed=1 accumulator=3853890695891366752
cpu_hog alive elapsed=2 accumulator=2052502631246739177
cpu_hog alive elapsed=3 accumulator=3747542254937611419
cpu_hog alive elapsed=4 accumulator=3569131114005427723
cpu_hog alive elapsed=5 accumulator=14262534237190810810
cpu_hog alive elapsed=6 accumulator=13503008120158151363
cpu_hog alive elapsed=7 accumulator=15226199755963252479
cpu_hog alive elapsed=8 accumulator=1200427456992044821
cpu_hog alive elapsed=9 accumulator=10509867372172427228
cpu_hog alive elapsed=10 accumulator=14679383370081987508
cpu_hog done duration=10 accumulator=14679383370081987508
```

COMMANDS

```
sudo ./engine start high rootfs-base ./cpu_hog --nice -5
sudo ./engine start low rootfs-base ./cpu_hog --nice 10
sudo ./engine logs high sudo ./engine logs low
```

8.Screenshot 8 — Clean teardown

Description:

All processes stopped and kernel module unloaded cleanly.

```

jyothika@os-jackfruit:~/OS-Jackfruit/boilerplate$ sudo pkill engine
ps aux | grep engine
sudo rmmod monitor
sudo dmesg | tail
root      3763   0.0  0.1 19188 3728 pts/1    D+   15:38   0:00 sudo ./engine supervisor rootfs-base
heemadh+  6574   0.0  0.0 6252 2008 pts/4      S+   16:07   0:00 grep engine
[ 1884.086471] [ 3790] 1000 3790 2247    410    0    410    0 61440    576    0 bash
[ 1884.086472] [ 4384]  0 4384  431     8    0     8    0 40960    0    0 sh
[ 1884.086474] [ 4411]  0 4411  431    24    0    24    0 40960    0    0 sh
[ 1884.086476] [ 4452]  0 4452  431     8    0     8    0 45056    0    0 sh
[ 1884.086477] [ 4560] 1000 4560 2247   502   96   406    0 57344   448    0 bash
[ 1884.086479] [ 5834] 1000 5834 2247   465   64   401    0 57344   512    0 bash
[ 1884.086480] [ 6476]  0 6476 856694 431044 431044    0    0 6897664 424384    0 memory_hog
[ 1884.086482] oom-kill:constraint=CONSTRAINT_NONE,nodemask=(null),cpuset=lightdm.service,mems_allowed=0,global_oom,task_memcg=/user.slice/user-1000.slice/session-2.scope,task=memory_hog,pid=6476,uid=0
[ 1884.086499] Out of memory: Killed process 6476 (memory_hog) total-vm:3426776kB, anon-rss:1724176kB, file-rss:0kB, shmem-rss:0kB, UID=0 pgtables:6736kB oom_score_adj:0
[ 1937.753653] [container_monitor] Module unloaded.

```

COMMANDS

sudo pkill engine ps

aux | grep engine

sudo rmmod

monitor sudo dmesg

| tail